A Reproducible and Explainable Machine Learning Framework for Frailty Risk Stratification Using Integrated Health and Social Determinants

Arpana Kumari
Master's in Computer Application
Lovely Professional University
ksony1524@gmail.com

***Abstract*—The aging global population requires rapid, accurate clinical triaging to identify vulnerable older adults before severe physical decline occurs. While machine learning improves upon manual frailty screening, it introduces a severe risk of opaque decision-making in clinical settings. This study evaluates 7,488 patient records to quantify the trade-off between clinical interpretability and predictive power. We compared a simple Logistic Regression model against an XGBoost model. Logistic Regression was easy to understand but missed 16% of frail patients. XGBoost was highly accurate at 98.80%. However, when we audited it using SHAP (SHapley Additive exPlanations), we found a hidden bias: the model mathematically penalized patients who had fewer years of education. This shows that while complex models are good at predicting frailty, they are not safe to use autonomously. We must use Explainable AI (XAI) and human oversight to stop the automation of healthcare inequalities.**

***Index Terms*—Frailty prediction, Explainable AI, Machine Learning, SHAP, Healthcare AI, XGBoost**

I. Introduction

Frailty is a decline in physical and mental health that makes older adults highly vulnerable to illness and injury. As healthcare systems manage an aging population, doctors need reliable, scalable tools to identify frail patients early for preventive care. Manual checking takes too long and often misses the complex ways that lifestyle, memory, and physical health interact.

Machine learning can solve this by looking at many health factors at once to flag at-risk patients during routine clinical assessments. But there is a major safety issue. Highly accurate models, like Gradient Boosted Trees, are "black boxes"—we cannot see how they get their answers. If a model denies a patient care based on their income instead of their physical health, it creates a systemic bias. This paper looks at how to use accurate models safely in clinical decision support systems by forcing transparency using XAI.

II. Methodology

A. Dataset and Preprocessing

We used a dataset of 7,488 patient records covering 14 different demographic, clinical, and lifestyle variables. The target variable for frailty status was perfectly balanced across three categories: Healthy, At-risk, and Frail. This natural balance meant we did not need to use synthetic oversampling techniques.

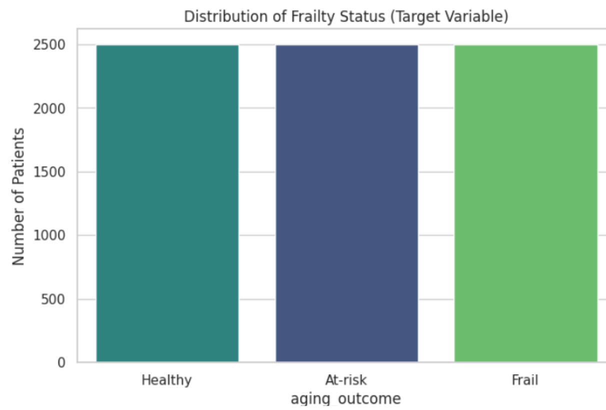

Fig. 1. Distribution of the target variable showing equal representation across Healthy, At-risk, and Frail categories.

Handling the data required making practical choices. The alcohol consumption feature was missing 53.5% of its data. Instead of dropping the column or guessing the missing numbers with averages, we labeled the missing data as "Unknown". This safely keeps the context of patients who chose not to report their habits. We then standardized the continuous numbers and used one-hot encoding for categorical text data.

B. Model Selection

We tested four different models to compare simple transparency against complex accuracy:

- **Logistic Regression:** A basic linear model used as our transparent baseline.
- **Random Forest:** An ensemble method used to capture non-linear relationships.
- **Neural Network (MLP):** A deep learning baseline.
- **XGBoost:** A gradient boosting framework chosen as our main champion model due to its high performance on structured data.

III. Results

The performance metrics showed a clear divide between simple models and complex ones.

The Logistic Regression baseline reached an 88.45% overall accuracy. However, in a medical setting, overall accuracy is not enough. The model had a Recall of 0.84 for the 'Frail' class. This means it had a 16% False Negative rate, missing nearly one in six highly vulnerable patients.

On the other hand, the XGBoost model achieved a 98.80% accuracy and a 0.9996 ROC-AUC score. Most importantly, it pushed the Recall for both frail and at-risk patients to 0.98, drastically reducing the number of missed diagnoses. To prove this was not a lucky test split, we ran a 5-Fold Stratified Cross-Validation, which returned a very stable mean accuracy of 98.42%.

TABLE I
COMPARISON OF MODEL PERFORMANCE

Model	Accuracy	ROC-AUC
Logistic Regression	0.8845	0.9705
Random Forest	0.9753	0.9989
Neural Network	0.9786	0.9946
XGBoost	0.9880	0.9996

IV. DISCUSSION: THE SHAP AUDIT

XGBoost fixed the problem of missing vulnerable patients, but deploying a black-box model in healthcare is risky. To see exactly how the model was making its decisions, we audited it using SHAP.

A. Medical Validation

The global SHAP summary showed that the model correctly learned basic medical logic. Memory score was the most important feature driving the predictions, followed by chronic conditions and age. This proves the model understands that cognitive decline is a major warning sign for physical frailty.

B. Algorithmic Bias

However, the audit also uncovered a major ethical problem. Education years was the second most important feature in the whole model. It outweighed direct physical health markers like BMI, blood pressure, and diet quality.

The SHAP beeswarm plot clearly showed that the model mathematically penalized patients who had fewer years of education, automatically inflating their frailty risk score regardless of how physically healthy they were. If a clinic used this algorithm without checking it, the system would unfairly judge patients based on their social status.

V. CONCLUSION

Highly accurate models like XGBoost are needed to properly predict frailty, as simpler models miss too many at-risk patients. However, this project proves that predictive accuracy is not the only thing that matters. Without audits, complex models will take shortcuts and use social biases, like education or income, to make medical decisions. To safely use AI in clinical decision support systems, predictive models must be paired with clear XAI dashboards. This ensures that doctors always have the final say and can override the AI if it is making biased choices.

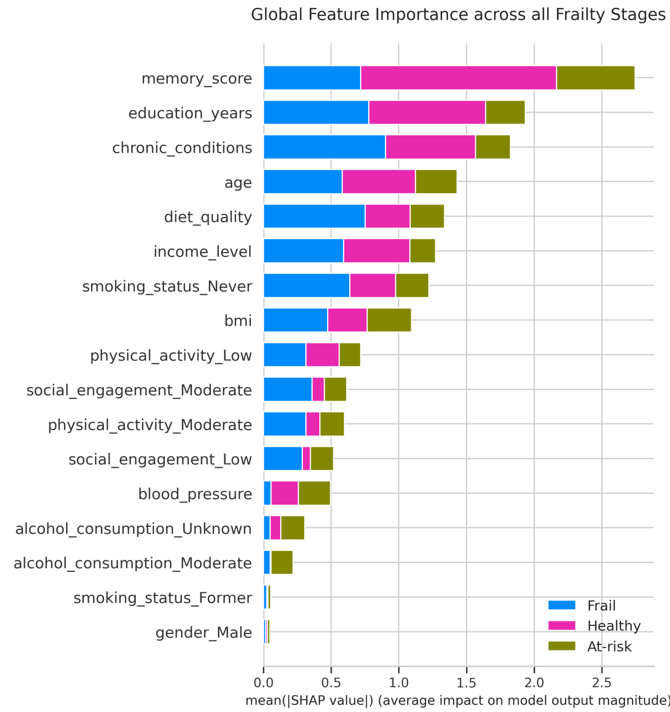

Fig. 2. Global Feature Importance showing Memory Score and Education Years as the top predictive drivers.

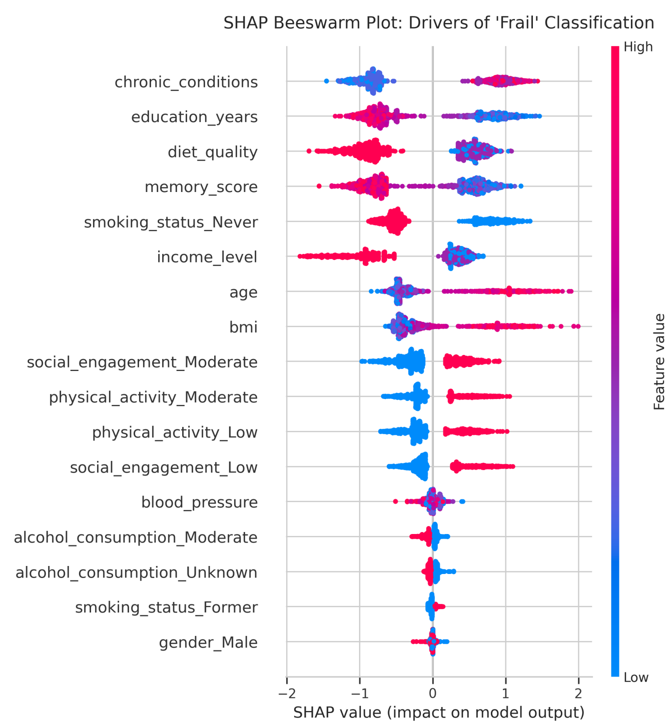

Fig. 3. SHAP Beeswarm Plot for the 'Frail' class. Blue dots on the right side of the education axis show how lower education mathematically increases the frailty risk score.

REFERENCES

[1] L. P. Fried et al., ”Frailty in older adults: evidence for a phenotype,” Journal of Gerontology, 2001.

[2] S. M. Lundberg and S.-I. Lee, ”A Unified Approach to Interpreting Model Predictions,” NeurIPS, 2017.

[3] T. Chen and C. Guestrin, ”XGBoost: A Scalable Tree Boosting System,” KDD, 2016.

[4] F. Pedregosa et al., ”Scikit-learn: Machine Learning in Python,” JMLR, 2011.